

Salah-Eddine Rizki

PhD Candidate in Artificial Intelligence

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Summary

PhD candidate working on graph representation learning and geometric deep learning for molecular modeling and machine-learning interatomic potentials, with broader interests in generative modeling and explainable AI.

Education

PhD	CY Cergy Paris University • ETIS Laboratory , Artificial Intelligence	Cergy, France
	- Research direction: explainable graph diffusion models. - Advisor: Vassilis Christophides.	2025 – 2029
M.Sc.	CY Cergy Paris University , Data Science and Machine Learning	Cergy, France
	GPA: 3.35/4.00	2024 – 2025
M.Eng.	ENSEA , Signal Processing and Artificial Intelligence	Cergy, France
	GPA: 3.55/4.00	2022 – 2025

Research

ETIS Laboratory • ENSEA • CY Cergy Paris University , PhD Researcher	Cergy, France
Research on graph representation learning, geometric deep learning, and explainable generative models, with current applications in molecular modeling and machine-learning interatomic potentials.	2025 – present
- Investigating explainable graph diffusion models for structured scientific data. - Studying molecular representations and interatomic-potential learning under geometric constraints.	
CY Cergy Paris University , Master's Thesis Researcher	Cergy, France
Conducted research on synthetic multivariate time-series generation from real data.	2025 – 2025

Experience

Betclic Group , Data Scientist Intern • Recommendation Systems	Bordeaux, France
Investigated sequential recommendation and foundation-model approaches for large-scale interaction data.	Apr 2025 – Oct 2025
	7 months
- Reviewed recent recommendation-system literature and implemented a BERT4Rec-based modeling pipeline. - Worked with more than 20 million interactions, 4,000 games, and 500,000 users. - Improved offline ranking metrics by 2.5× and contributed to a 15% increase in engagement. - Developed reproducible workflows with Docker, AWS SageMaker, Airflow, and Jenkins.	
AEGLE , Machine Learning Engineer Intern	Paris, France
Designed and evaluated anomaly-detection methods on operational data.	Apr 2024 – Sept 2024
- Trained and evaluated models on more than 50,000 records, reaching 94% accuracy. - Reduced false positives by 15% and improved F1 score by 10% through feature and model analysis.	6 months

OCP Group, Data Scientist Assistant Intern

Built analytical tools for monitoring industrial operations.

- Developed Python and SQL dashboards tracking 20 KPIs across five production units.

Casablanca, Morocco

Aug 2023 – Oct 2023

3 months

Publications

Counterfactual Explanations for Equivariant Graph Neural Networks: CE-EGNN for Molecular Property Regression

Jan 2026

Counterfactual explanation method for equivariant graph neural networks in molecular property regression.

Salah-Eddine Rizki, Fatia Lekbour, Guillaume Renton

edbticdt2026.github.io/workshop_papers/XAI4Science-paper7.pdf (EDBT/ICDT 2026 Workshops)

Presentations

Poster presentation • Learning on Graphs Paris Meetup 2026

Paris, France

Presented research as part of the LoG Paris research community meetup.

June 2026

Accelerating Science with AI

Mar 2026

Research talk with Vassilis Christophides.

Projects

Message-Passing GNNs from Scratch

Transparent implementations of GCN and GAT in pure PyTorch, without PyTorch Geometric, with controlled classification, regression, and oversmoothing experiments.

- Sparse edge-index message passing
- Reproducible GCN/GAT comparisons

Financial Time-Series Transformer

Transformer-based modeling experiments for financial time series.

Fraud Detection

Machine-learning pipeline for fraud detection and model evaluation.

Skills

Research areas: Graph representation learning, geometric deep learning, graph generative models, explainable AI, molecular machine learning, MLIPs

Machine learning: PyTorch, deep learning, graph neural networks, transformers, recommendation systems, time-series modeling

Engineering: Python, SQL, Docker, AWS SageMaker, Airflow, Jenkins, Git